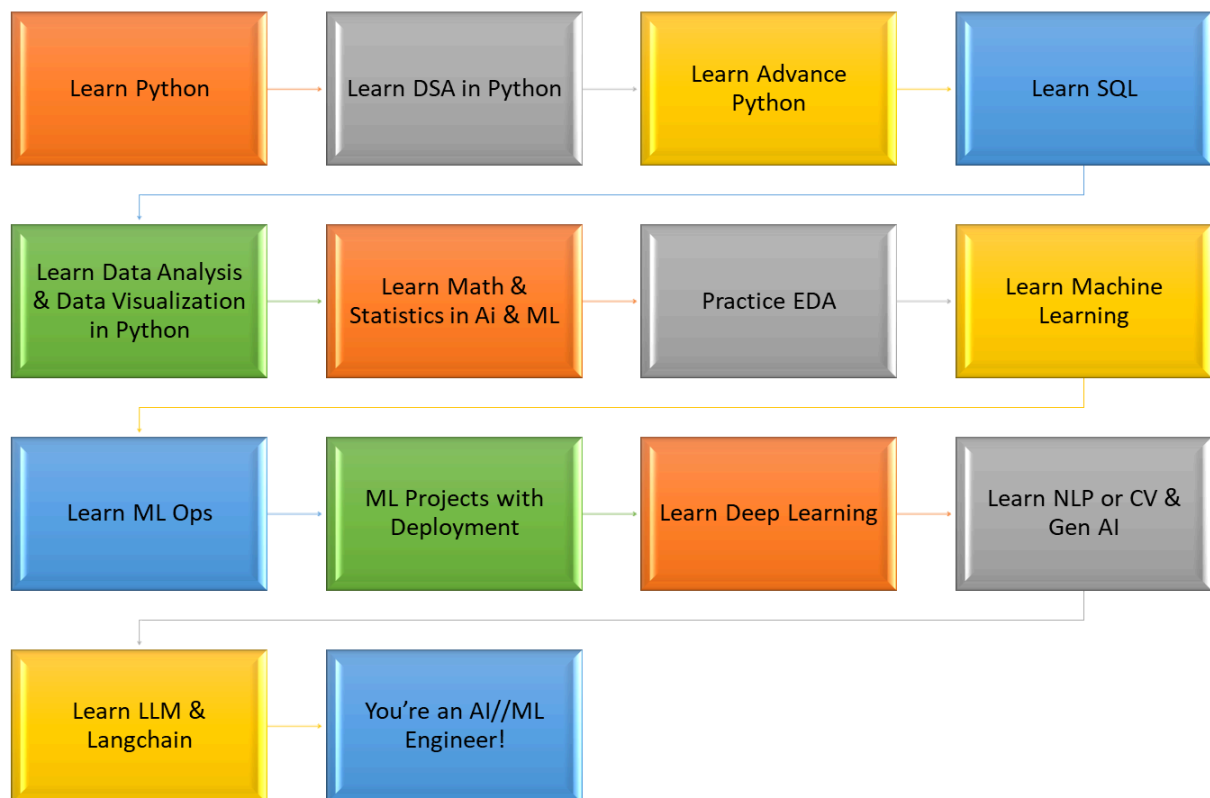


Detailed Roadmap & Free Resources to Learn AI

This roadmap is designed for complete beginners to acquire AI/ML Engineer skills, even without any prior coding or computer science knowledge. It covers all technical skills with free learning resources provided for each.

Expected Duration: 8 to 10 months

How to Become an AI/ML Engineer



Step 1 - Learn Python

Topics you need to cover in Python

- Variables, Numbers, Strings
- Lists, Dictionaries, Sets, Tuples
- If condition, for loop
- Functions, Lambda Functions
- Modules (pip install)
- Read, Write files

- Exception handling
- Classes, Objects

YouTube Videos to learn Python

- https://youtube.com/playlist?list=PL-osiE80TeTt2d9bfVyTiXJA-UTHn6WwU&si=9ING_XKiJKrSYDgR
- https://youtube.com/playlist?list=PLu0W_9lI9agwh1XjRt242xIpHhPT2llg&si=uY9q2ceGXl6bmQl4

Free Courses to learn Python

- <https://www.edx.org/learn/python/harvard-university-cs50-s-introduction-to-programming-with-python>
- <https://www.geeksforgeeks.org/courses/python-course-certification-free>
- <https://www.freecodecamp.org/learn/scientific-computing-with-python/>

Exercises to learn Python

- <https://github.com/codebasics/py/tree/master/Basics/Exercise>

Step 2 - Learn DSA in Python

Topics you need to cover for DSA in Python

- Data structures basics, Big O notation
- Data structures: Arrays, Linked List, Hash Table, Stack, Queue
- Data structures: Tree, Graph
- Algorithms: Binary search, Bubble sort, quick sort, merge sort
- Recursion

YouTube Videos to learn DSA (You can choose anyone from the below playlists)

- https://www.youtube.com/playlist?list=PLeo1K3hjS3uu_n_a_MI_KktGTLYopZ12
- <https://www.youtube.com/watch?v=MLqHDsBOC4c&t=551s>
- <https://www.youtube.com/watch?v=pkYVOMU3MgA&t=2277s>
- <https://www.youtube.com/watch?v=mGJusQ7iBwc>

Step 3 - Advance Python

Topics you need to cover

- Inheritance
- Generators
- Iterators
- List Comprehensions
- Decorators
- Multithreading
- Multiprocessing

YouTube Videos to learn Advanced Python (You can choose anyone from the below playlists)

- <https://www.youtube.com/playlist?list=PLqnsIRFeH2UqLwzS0AwKDKLrpYBKzLBy2>
- <https://www.youtube.com/playlist?list=PLeo1K3hjS3uv5U-LmInucd7gqF-3ehlh0>

Step 4 - Learn SQL

SQL is extremely important for learning AI-related skills. It allows you to work with structured data stored in databases. As an AI Engineer, you need SQL to extract, manipulate, and analyze data from these databases.

YouTube videos to learn SQL

- <https://www.youtube.com/watch?v=noplGY1zJE0>
- <https://www.youtube.com/watch?v=2w-M4z-iQpk>

Free Courses to learn SQL

- <https://www.w3schools.com/sql/>
- <https://sqlbolt.com/>

Step 5 - Learn Data Analysis & Data Visualization in Python

For Data Analysis in Python, learn Numpy & Pandas

- a. YouTube videos to learn Numpy (You can choose anyone from the below playlists)
 - <https://www.youtube.com/playlist?list=PLeo1K3hjS3uset9zIVzJWqplaWBiacTEU>
 - <https://www.youtube.com/watch?v=ZaKzw9tULeM&list=PLjVLYmrlmjGfgBKkIFBkMNGG7qyRfo00W>
 - <https://youtube.com/playlist?list=PL9n0l8rSshSnragNblKDBsT8Xu3otp3jA&si=ltkXQ5eAegFYbkWk>
- b. YouTube videos to learn Pandas (You can choose anyone from the below playlists)
 - <https://youtube.com/playlist?list=PLeo1K3hjS3uuASpe-1LjfG5f14Bnozjwy&si=0HgEHxoGYWpbSPcS>
 - <https://youtube.com/playlist?list=PLjVLYmrlmjGdEE2jFpL71LsVH5QjDP5s4&si=K5UEPCKXoIFWusN8>

For Data Visualization in Python, learn Matplotlib & Seaborn

- a. YouTube videos to learn Matplotlib (You can choose anyone from the below playlists)
 - https://youtube.com/playlist?list=PLeo1K3hjS3uu4Lr8_kro2AqaO6CFYgKOI&si=ejpRnZcF1JQhJUtk

- https://youtube.com/playlist?list=PLjVLYmrlmjGcC0B_FP3bkJ-JIPkV5GuZR&si=klf5Fi7QaNyRZVqT
- b. YouTube videos to learn Seaborn
 - <https://youtube.com/playlist?list=PLjVLYmrlmjGfhqSO3rF4n02rrj9w2Ch2C&si=sUuG7RZj6c1VaX47>

Step 6 - Learn Maths & Statistics for Ai & ML

Topics you need to cover in Statistics & Probability for Data Science

- Basics: Descriptive vs inferential statistics, continuous vs discrete data, nominal vs ordinal data
- Linear Algebra: Vectors, Matrices, Eigenvalues and Eigenvectors
- Calculus: Basics of integral and differential calculus
- Basic plots: Histograms, pie charts, bar charts, scatter plots, etc.
- Measures of central tendency: mean, median, mode
- Measures of dispersion: variance, standard deviation
- Probability basics
- Distributions: Normal distribution
- Correlation and covariance
- Central limit theorem
- Hypothesis testing: p-value, confidence interval, type 1 vs type 2 error, Z test, t-test, ANOVA

YouTube Videos to learn Statistics & Probability for Ai & ML (You can choose anyone from the below playlists)

- <https://www.youtube.com/watch?v=LZzq1zSL1bs>
- <https://www.youtube.com/watch?v=Vfo5le26lhY>
- <https://www.youtube.com/watch?v=bLZ-LSsQMCC>

Now, after learning Python, DSA in Python, SQL, Data Analysis, Data Visualization using Python, & Statistics for AI & ML, you can use all these to do Exploratory Data Analysis, which is the first step in building a model.

Step 7 - Practice Exploratory Data Analysis

Watch anyone of the videos below to understand how to do EDA -

- https://www.youtube.com/watch?v=fFi_TBw27is&list=PL9n0l8rSshSnZHFChXIG5Q9f0UN8O-UYX&index=3

- [youtube.com/watch?v=fHFOANOHwh8](https://www.youtube.com/watch?v=fHFOANOHwh8)

Use Kaggle to find Datasets for Exploratory Data Analysis (Practice EDA on at least 5 datasets) - <https://www.kaggle.com/datasets?search=Exploratory+Data+Analysis>

Step 8 - Learn Machine Learning (Most important)

Topics to cover:

Machine Learning: Preprocessing

- Handling NA values, outlier treatment, data normalization
- One hot encoding, label encoding
- Feature engineering
- Train test split
- Cross-validation

Machine Learning: Model Building

- Types of ML: Supervised, Unsupervised
- Supervised: Regression vs Classification
- Linear models
 - Linear regression, logistic regression
 - Gradient descent
- Nonlinear models (tree-based models)
 - Decision tree
 - Random forest
 - XGBoost
- Model evaluation
 - Regression: Mean Squared Error, Mean Absolute Error, MAPE
 - Classification: Accuracy, Precision-Recall, F1 Score, ROC Curve, Confusion matrix
 - Hyperparameter tuning: GridSearchCV, RandomSearchCV
- Unsupervised: K means, Hierarchical clustering, Dimensionality reduction (PCA)

Youtube Videos to learn Machine Learning

- <https://youtube.com/playlist?list=PLeo1K3hjS3uvCeTYTeyfe0-rN5r8zn9rw&si=WZA6lTnDR2-YPEfj>
- <https://www.youtube.com/watch?v=NWONeJKn6kc>
- <https://www.edx.org/learn/machine-learning/harvard-university-data-science-machine-learning>

Step 8 - Learn ML Ops

Topics to cover:

- What is API?
- FastAPI for Python server development
- DevOps Fundamentals: CI/CD pipelines, containerization (Docker, Kubernetes)
- Familiarity with at least one cloud platform (AWS, Azure etc.)

Youtube Videos to learn

- <https://www.youtube.com/watch?v=e1rl1DYfPy4>
- <https://www.youtube.com/watch?v=Wrl1JjhTt1Xg>
- <https://www.youtube.com/watch?v=pTFZFxd4hOI>

Step 9 - Machine Learning Projects with Deployment

Do end-to-end ML projects covering both Regression & Classification

For Regression

<https://youtube.com/playlist?list=PLeo1K3hjS3uu7clOTwsp94PcHbzqpAdg&si=tQNxP-xopMIEa0CI>

For Classification

<https://youtube.com/playlist?list=PLeo1K3hjS3uvaRHZLI-jLovljBP14QTXc&si=5E2O2XT5VaccMWK-a>

ML Projects

- https://youtube.com/playlist?list=PLKnIA16_RmvY5eP91BGPao0vXUYmldtfPQ&si=FLiysiVRxYp_xkgt
- https://youtube.com/playlist?list=PLfFghEzKvmjvuSA67LszN1dZ-Dd_pkus6&si=hb4jc_0sr4W80GIn

You can also find ML projects on the Kaggle Platform - <https://www.kaggle.com/>

Step 10 - Learn Deep Learning

Topics to Cover

- What is a neural network?
- Building multilayer perceptron
- Special neural network architectures
 - Convolutional neural network (CNN)
 - Sequence models: RNN, LSTM

YouTube Playlists

- https://youtube.com/playlist?list=PLeo1K3hjS3uu7CxAacxVndI4bE_o3BDtO&si=OJA g9Mfb-hfUapM
- https://youtube.com/playlist?list=PLOrU905yPYXJsJSHJsiE779KfcrRCgz4v&si=_yp3 K8IOEpAAqcS5
- https://youtube.com/playlist?list=PLKnIA16_RmvYuZauWaPIRTC54KxSNLtNn&si=zd IKOR8dhs26vJxj

Projects

- <https://youtube.com/playlist?list=PLeo1K3hjS3utJFNGyBplvjWgSDY0eOE8S&si=aP Z4if0tzQYI8fok>
- <https://youtube.com/playlist?list=PLfFghEzKVmjvyT7wdfDJSHhS0FdF8jd-U&si=DfB U6k28idQuG5iU>

Step 11 - Learn NLP or Computer Vision & GenAI

Choose a specialized track, which is either NLP or Computer vision. No need to learn both.

NLP Youtube Playlist -

<https://youtube.com/playlist?list=PLeo1K3hjS3uuvuAXhYjV2IMEShq2UYSwX&si=59s20mkb B5-7RYwP>

CV Youtube playlist -

<https://youtube.com/playlist?list=PLzWRmD0Vi2KVsrCqA4VnztE4t71KnTnP5&si=f0Rcf0dib RCsUMmS>

Step 12 - Learn LLM & Langchain

Topics to Cover

- What are LLM, Vector database, and Embeddings?
- RAG (Retrieval Augmented Generation)
- Langchain framework

YouTube Videos to learn LLM & Langchain

- https://www.youtube.com/playlist?list=PLeo1K3hjS3uu0N_0W6giDXzZlcB07Ng_F
- https://youtube.com/playlist?list=PLZoTAEELRMXVORE4VF7WQ_fAI0L1Gljtar&si=HM z3PTaW1oO4r7ex
- <youtube.com/watch?v=aWKrL4z5H6w>
- <https://www.youtube.com/watch?v=xZDB1naRUIk>

Congratulations, You're an AI/ML Engineer!

Free Certifications on AI-related Skills

- <https://www.sololearn.com/en/learn>
- <https://www.simplilearn.com/skillup-free-online-courses>
- <https://www.mygreatlearning.com/academy>
- <https://aimagazine.com/top10/top-10-ai-certifications> (few of them are paid)